Blockchain solutions for incentivization of market-linked upskilling in labor markets

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***Abstract*—Blockchain technology has proved to be useful for basic applications such as peer-to-peer electronic cash systems i.e., Bitcoin [1]. Since this milestone, the technology has evolved to support advanced applications which require more elaborate feature sets. Ethereum [2] is an example of one such form of blockchain technology that allows public deployment of capture-resistant, permissionless contracts, resulting in a platform on which interested parties may attempt to design systems that better address systemic social challenges.**

This paper proposes blockchain solutions for incentivization of market-linked upskilling in South African labor markets with intent to address persistent unemployment in Stellenbosch, South Africa - a problem that is hypothesized to be due in part to the cold-start problem of underprivileged individuals in a dual-speed economy.

The incentive scheme comprises three blockchain solutions: incentive tokens disbursed to learners on completion of course milestones, verified credentials assigned to learners for completion of courses, and HyperCerts awarded to educators for the effective upskilling of learners via courses. The solution(s) is investigated in distinct parts and as a whole by: architecting the components of the solution; simulating the emergent scheme as a sum of the solutions; and surveying relevant stakeholders on the perceived efficacy of such a solution with respect to resolving local unemployment in the town of Stellenbosch.

***Index Terms*—blockchain, tokenomics, HyperCert, Ethereum, incentivisation, unemployment**

I. Introduction

Blockchain technology has proved valuable for applications ranging from peer-to-peer electronic cash systems such as Bitcoin [1] to general-purpose computing platforms such as Ethereum [2]. Ethereum offers a Turing-complete scripting environment that enables developers to encode arbitrary rules for ownership, transaction formats, and state transitions. This potential has given rise to a large and growing class of decentralized applications (DApps), with recent surveys cataloguing thousands of deployed DApps across domains including finance, gaming, governance, and identity [3]. The emergence of a composable platform with fundamental properties of decentralization, persistence, and auditability, poses a strong case for the design of systems that address pressing systemic social challenges.

Realising ambitious application-layer systems on Ethereum has historically been constrained by the base layer's limited throughput and high transaction costs. A substantial body of recent work addresses these constraints through Layer 2 (L2) scaling architectures including state channels, sidechains, and, most prominently, optimistic and zero-knowledge rollups [4], [5]. Rollup-based L2s now offer orders-of-magnitude higher throughput while inheriting the security guarantees of the underlying L1, rendering them sufficiently mature to reliably host production-scale social and economic applications. It is recognized that are there still pressing technical challenges to be overcome by the Ethereum network (cite); however, the existing utility offered by the network is sufficient for the purposes of this research. The maturity of the network is confirmed by an accelerated interest in blockchain for social good. Sansone et al [6] identify reliability, transparency, decentralization, and accessibility as the properties through which blockchain can support social innovation, while earlier work documents proof-of-concept deployments spanning humanitarian aid, supply-chain transparency, digital identity, and philanthropy [7]. Much of this work nevertheless remains exploratory, with limited engagement with specific, deeply rooted socio-economic problems in a particular context.

South Africa presents one such context. Youth unemployment has persisted above forty-five percent for more than a decade and, in Stellenbosch in particular, is driven in significant part by a dual-speed economy in which underprivileged entrants face a structural cold-start problem: they cannot acquire market-relevant experience without initial access to the labour market, and cannot access the labour market without prior experience or at least highly relevant skills (mostly gated by experience in labour markets) [8]–[10]. Supply-side interventions targeting general employment skills alone have been shown to be insufficient; they must be coupled with demand-side incentives that connect learners, educators, and employers [8], [16]. It is reasonable to suggest that sophisticated incentives for employers to inform the upskilling of the unemployed may create that appropriate dynamic that is needed for impactful unemployment intervention. There are three benefits in such a dynamic that are immediately apparent: the unemployed learn skills that are highly relevant for employment; the employers are able to 'pre-train' interested individuals in the unemployed population; the employers are rewarded with financial incentive. However, the cold-start problem where unemployed individuals lack the means to

provide for themselves in the short-term still remains. The proposed solution to this is the disbursement of incentive tokens on completion of course milestones, which can be used to purchase basic goods at a network of pre-approved vendors, who exchange the tokens for ZAR stablecoin, and then ZAR. The ZAR stablecoins are either funded by government municipalities, or by Corporate Social Responsibility programs of the employer-educators.

While the merit of this hypothesis is subject to further social studies contextualised to Stellenbosch, the focus of this paper is on the technical feasibility of such a system, making the large assumption that the hypothesis is true in the context of this research.

This paper recognises the support for designing application-layer systems on Ethereum and the sufficient need for such systems in the South African labour market context and proposes a blockchain-based incentivization scheme for market-linked upskilling in the South African labour market, composed of three interrelated solutions: fungible incentive tokens disbursed to learners on the completion of course milestones; non-transferable, or soulbound, credential tokens assigned to learners for course completion, in accordance to the decentralized society proposal of Weyl et al. [11]; and HyperCerts [12] awarded to educators for the effective upskilling of learners. The combined ecosystem is designed using frameworks that treat incentive design, governance, and tokenomics as interdependent pillars of a token economy [13]–[15]. The remainder of the paper architects each component, simulates the resulting system to study its tokenomics and tune parameters, and surveys stakeholders on the perceived efficacy of such a scheme in addressing structural unemployment.

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TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
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[a]Sample of a Table footnote.

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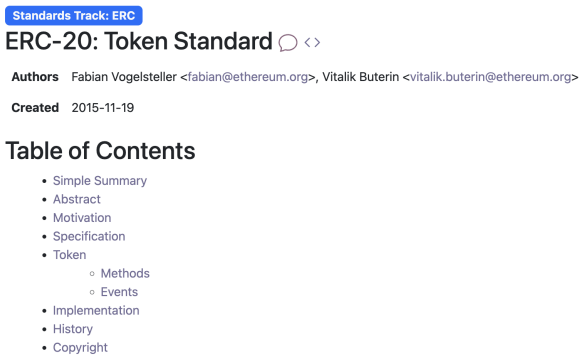

Standards Track: ERC

ERC-20: Token Standard

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Fig. 1. Example of a figure caption.

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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